

Admissibility Conditions: Valid System Description in Dimensional Human Field Theory (DHFT)

DHFT Technical Note: Admissibility Conditions

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Abstract

This document defines the admissibility conditions governing valid system description in Dimensional Human Field Theory (DHFT). The system is specified over invariant variables Load (L), Capacity (C), and Boundary (B).

The conditions defined here establish what constitutes a valid description at the minimal layer and constrain the introduction of additional variables, interpretations, or domain-specific constructs.

All admissible descriptions reduce to the invariant variable set and preserve the structural relations defined in DHFT.

I. Scope

This document defines the conditions under which a system description is considered valid within DHFT. It specifies the constraints governing admissibility at the minimal layer.

No interpretive, psychological, or domain-specific constructs are introduced.

II. Variable Reference

The system is defined over three invariant variables: Load (L), Capacity (C), and Boundary (B). These variables are fixed at the minimal layer and are not expanded or redefined within this document.

III. Admissibility Conditions

A system description is admissible if and only if it reduces to Load, Capacity, and Boundary and preserves the relations defined in DHFT.

No additional variables are permitted at the minimal layer.

All valid descriptions maintain consistency with the stability conditions, boundary-regulated flow, and causal structure defined in the system.

IV. Exclusion Criteria

A system description is not admissible if it introduces additional variables, substitutes interpretive constructs for structural relations, or alters the defined causal order.

Descriptions that depend on perception, narrative, intent, or domain-specific meaning are excluded at the minimal layer.

V. Invariance Requirement

All admissible system descriptions preserve the invariance of Load, Capacity, and Boundary. Variable meaning is not modified across contexts or applications.

All valid descriptions remain reducible to the same invariant structure.

VI. Constraint

The admissibility conditions defined here constrain all valid use of DHFT at the minimal layer. Any extension, application, or interpretation must remain consistent with these conditions.

VII. Statement

The admissibility conditions defined here establish the criteria for valid system description in DHFT. All admissible descriptions preserve variable invariance, structural relations, and causal order.

VIII. System Reference

The system defined in this document is specified in *Dimensional Human Field Theory (DHFT): Architectural Identification of a Closed Explanatory System*, DOI: 10.5281/zenodo.19075948.

All valid system descriptions reduce to Load, Capacity, and Boundary.

No additional variables are introduced at the minimal layer.

This document is part of the DHFT technical series.

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This work defines original structural models and associated terminology within Dimensional Human Field Theory (DHFT).

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